ENGINEERING PHYSICS | B.TECH 1ST YEAR

Wave Optics

Unit I – Interference, Diffraction & Polarization

SUBJECT	PROGRAMME	UNIT
Engineering Physics	B.Tech – Semester I/II	Unit I (Complete)

CONTENTS

1 Introduction to Wave Optics

Nature of Light · Wave Theory · Historical Background

1.1 What is Wave Optics?

Wave Optics, also known as Physical Optics, is the branch of optics that treats light as an electromagnetic wave rather than a ray. While Geometric (Ray) Optics explains phenomena such as reflection and refraction using straight-line propagation, it fails to account for effects like interference, diffraction, and polarization that can only be explained by the wave nature of light.

Light is a transverse electromagnetic wave with an electric field vector **E** and a magnetic field vector **B** oscillating perpendicular to each other and to the direction of propagation. In vacuum, light travels at speed $c = 3 \times 10^8$ m s^{-1}.

1.2 Historical Development

Christian Huygens (1678) proposed the Wave Theory of light, explaining reflection and refraction by treating each point on a wavefront as a source of secondary wavelets. **Thomas Young (1801)** demonstrated the wave nature conclusively through his double-slit experiment. **James Clerk Maxwell (1864)** unified the theory by showing light is an electromagnetic wave. **Albert Einstein (1905)** introduced the photon concept (particle nature), giving rise to wave-particle duality.

ⓘ **KEY INSIGHT**

Wave Optics is essential to understand phenomena that Geometric Optics cannot explain: Interference (thin films, Newton's rings), Diffraction (bending around edges), and Polarization (transverse wave property).

1.3 Electromagnetic Nature of Light

Light is characterized by: wavelength (λ), frequency (ν), amplitude (A), phase (ϕ), and direction of polarization. The relationship between these is:

Speed: $c = \nu\lambda$

Wave number: $k = 2\pi / \lambda$

Angular frequency: $\omega = 2\pi\nu$

Wave equation: $y = A \sin(\omega t - kx)$

Intensity: $I \propto A^2$

1.4 Visible Spectrum

Colour	Wavelength Range (nm)	Frequency Range (THz)
Violet	380 – 450	666 – 789
Blue	450 – 495	606 – 666
Green	495 – 570	526 – 606
Yellow	570 – 590	508 – 526
Orange	590 – 620	484 – 508
Red	620 – 750	400 – 484

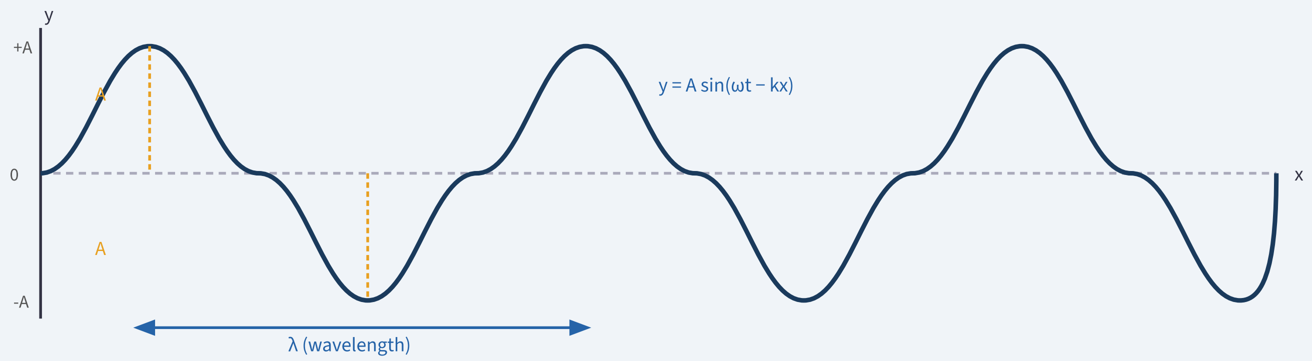


FIG. 1.1 — TRANSVERSE WAVE REPRESENTATION

$y = A \sin(\omega t - kx)$ showing amplitude A and wavelength λ

2 Interference Overview

Coherence · Conditions · Constructive & Destructive

2.1 Definition of Interference

Interference is the phenomenon in which two or more light waves superpose to form a resultant wave of greater, lesser, or the same amplitude. It arises from the wave nature of light and is observable when the superposing waves have a fixed phase relationship (coherence).

2.2 Coherence

For sustained interference, the two sources must be *coherent*: they must emit waves of the same frequency and maintain a constant phase difference. Two types of coherence are important:

TEMPORAL COHERENCE

Related to the monochromaticity (purity) of the source. A perfectly monochromatic source has infinite temporal coherence.

SPATIAL COHERENCE

Related to the lateral extent of the source. A point source has perfect spatial coherence.

2.3 Conditions for Interference

1. The two sources must be coherent (same frequency, constant phase difference).
2. The waves must have the same or nearly the same amplitude for good contrast.
3. The path difference must be within the coherence length of the source.
4. For polarized light, the vibration directions must be parallel (same plane of polarization).
5. The sources should be closely spaced (separation much smaller than distance to screen).

2.4 Types of Interference

Type	Path Difference (Δ)	Phase Difference (δ)	Result
Constructive	$n\lambda$ ($n = 0, 1, 2, \ldots$)	$2n\pi$	Bright fringe (maximum intensity)
Destructive	$(2n+1)\lambda/2$	$(2n+1)\pi$	Dark fringe (minimum intensity)

3

Principle of Superposition

Resultant Amplitude · Phase Analysis · Intensity

3.1 Statement

The Principle of Superposition states: When two or more waves meet at a point simultaneously, the resultant displacement at that point is the algebraic sum of the individual displacements produced by each wave independently.

> 💡 **IMPORTANT**
>
> The superposition principle holds only in linear media. After superposition, each wave continues to propagate independently as if the other did not exist.

3.2 Mathematical Derivation

Let two waves of the same frequency and amplitude be:

$$y_1 = A \sin(\omega t)$$

$$y_2 = A \sin(\omega t + \delta)$$

where δ = phase difference between the two waves

By superposition, the resultant displacement is:

$$y = y_1 + y_2 = A \sin(\omega t) + A \sin(\omega t + \delta)$$

$$y = A [\sin(\omega t) + \sin(\omega t + \delta)]$$

Using: $\sin P + \sin Q = 2 \sin((P+Q)/2) \cos((P-Q)/2)$

$$y = 2A \cos(\delta/2) \cdot \sin(\omega t + \delta/2)$$

$$y = R \sin(\omega t + \delta/2)$$

Resultant amplitude: $R = 2A \cos(\delta/2)$

3.3 Resultant Intensity

$$I \propto R^2 = 4A^2 \cos^2(\delta/2)$$

$$I = 4I_0 \cos^2(\delta/2) \quad [\text{where } I_0 = A^2]$$

For $\delta = 0$ (maxima): $I_{max} = 4I_0$

For $\delta = \pi$ (minima): $I_{min} = 0$

Visibility: $V = (I_{max} - I_{min}) / (I_{max} + I_{min})$

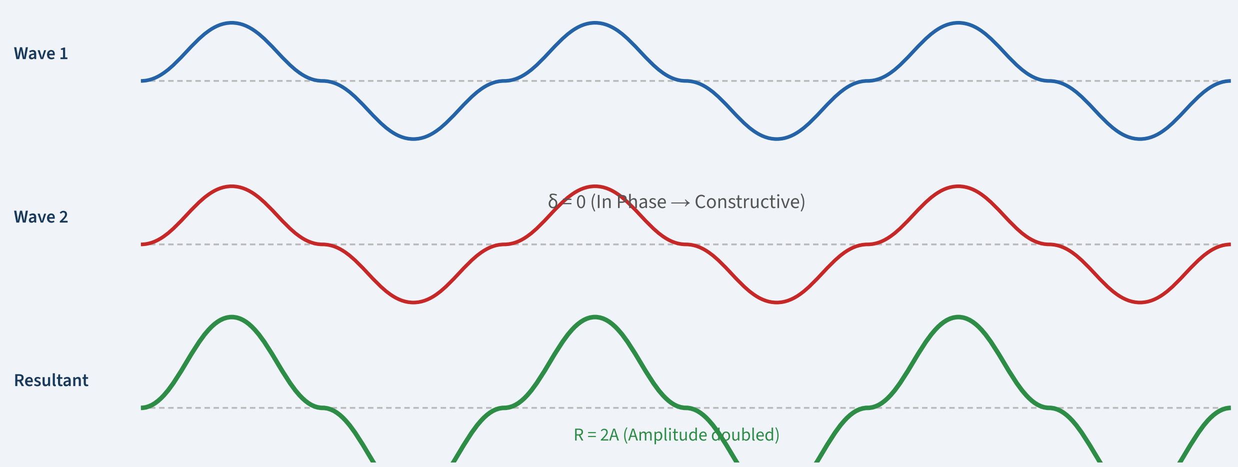


FIG. 3.1 — SUPERPOSITION OF TWO COHERENT WAVES (Δ = 0)

When two waves are in phase, resultant amplitude $R = 2A$ and intensity $I = 4I_0$

4

Interference of Light – Young's Experiment

Double Slit · Fringe Width · Path Difference Derivation

4.1 Young's Double Slit Experiment

Thomas Young (1801) first demonstrated interference of light by allowing monochromatic light to pass through two closely spaced narrow slits S_1 and S_2 (separated by distance d) and observing alternate bright and dark fringes on a screen at distance D.

4.2 Derivation of Fringe Width

Let S_1 and S_2 be two coherent sources separated by distance d. Screen is at distance D. Point P is at height y from centre O.

1. Path difference: $\Delta = S_2P - S_1P$
   Using geometry: $\Delta = y \cdot d / D$ (for small angles)
2. For bright fringe (maxima): $\Delta = n\lambda \Rightarrow y_n = n\lambda D/d$
3. For dark fringe (minima): $\Delta = (2n-1)\lambda/2 \Rightarrow y_n = (2n-1)\lambda D/(2d)$
4. Fringe width β = distance between consecutive bright (or dark) fringes:
   $\beta = y_{n+1} - y_n = \lambda D/d$

Path diff.: $\Delta = yd/D$

Bright fringes: $y_n = n\lambda D/d \quad (n = 0, \pm1, \pm2, \ldots)$

Dark fringes: $y_n = (2n-1)\lambda D / (2d) \quad (n = 1, 2, 3, \ldots)$

Fringe width: $\beta = \lambda D/d$

Angular fringe: $\theta = \lambda/d$

4.3 Intensity Distribution

$$I = 4I_0 \cos^2(\delta/2)$$

where: $\delta = (2\pi/\lambda) \times \Delta = (2\pi/\lambda) \times yd/D$

$I_{max} = 4I_0$ (at $\delta = 2n\pi$)

$I_{min} = 0$ (at $\delta = (2n+1)\pi$)

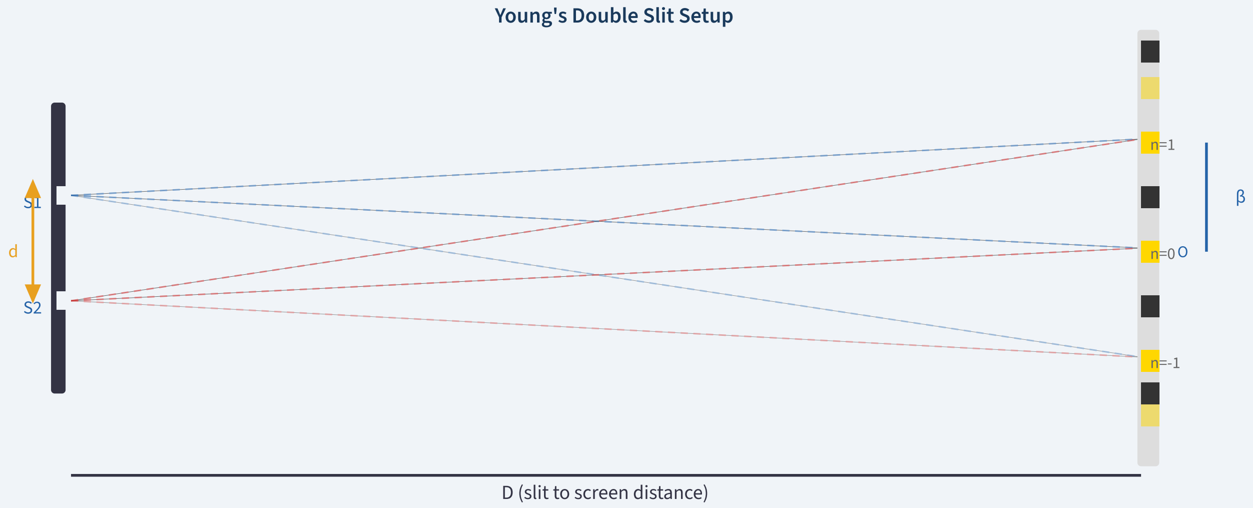


FIG. 4.1 — YOUNG'S DOUBLE SLIT INTERFERENCE

Fringe width $\beta = \lambda D/d$ | Bright fringes (gold) alternate with dark fringes (black)

5 Thin Film Interference

Reflection Geometry · Optical Path · Newton's Rings

5.1 Introduction to Thin Films

When light falls on a thin transparent film (such as a soap bubble or oil film on water), some light is reflected from the top surface and some from the bottom surface. These two reflected beams are derived from the same source and hence are coherent; they interfere to produce coloured patterns.

5.2 Derivation – Reflected Light (Stokes Relations)

Consider a thin film of thickness t, refractive index μ, with light of wavelength λ incident at angle θ_r (angle of refraction inside film).

1. Ray 1 is reflected at upper surface (air–film boundary, denser medium): phase change of π (equivalent path change of λ/2).
2. Ray 2 travels through film, reflects at lower surface (film–air boundary, rarer medium): no phase change. Then refracts back out.
3. Geometric path difference: $2\mu t \cos \theta_r$
4. Accounting for phase reversal at top surface, effective path difference = $2\mu t \cos \theta_r - \lambda/2$
5. For constructive interference (bright): $2\mu t \cos \theta_r - \lambda/2 = n\lambda$
   $\Rightarrow 2\mu t \cos \theta_r = (2n+1)\lambda/2$
6. For destructive interference (dark): $2\mu t \cos \theta_r - \lambda/2 = (2n-1)\lambda/2$
   $\Rightarrow 2\mu t \cos \theta_r = n\lambda$

Optical path diff.: $\Delta = 2\mu t \cos\theta_r$

Effective (reflected): $\Delta_{eff} = 2\mu t \cos\theta_r - \lambda/2$

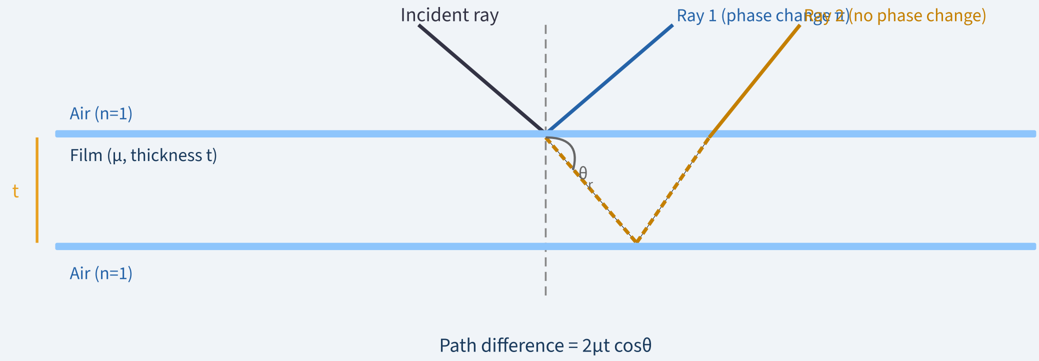
Bright (reflected): $2\mu t \cos\theta_r = (2n+1)\lambda/2 \quad n = 0, 1, 2, \ldots$

Dark (reflected): $2\mu t \cos\theta_r = n\lambda \quad n = 0, 1, 2, \ldots$

Normal incidence: $\theta_r = 0, \cos\theta_r = 1$

5.3 Colours in Thin Films

When white light falls on a thin film of non-uniform thickness, different wavelengths satisfy the bright-fringe condition at different thicknesses, producing beautiful colours. The condition $2\mu t \cos\theta_r = (2n+1)\lambda/2$ shows that which wavelength appears bright at a given point depends on the local thickness t. As t varies continuously (e.g., in a soap bubble), the film displays a spectrum of colours.



$_r$ Effective path diff. = 2μt cosθ_r − λ/2 (due to phase reversal at top) Fig 5.1 Thin Film Reflection Geometry

FIG. 5.1 — THIN FILM INTERFERENCE (REFLECTION)

Ray 1 suffers phase reversal at air-to-film boundary; Ray 2 travels extra path 2μt cosθ_r

5.4 Applications of Thin Film Interference

Application	Principle Used	Example
Anti-reflection coatings	Destructive interference minimises reflection	Camera lenses, spectacle lenses
High-reflectance coatings	Constructive interference maximises reflection	Laser mirrors
Soap bubbles / oil films	Colour-selective interference with white light	Decorative iridescence
Interferometry	Fringe shift measures thickness / flatness	Optical testing of lenses
Filters (bandpass)	Only selected wavelength passes	Spectroscopy instruments

6 Newton's Rings

Experimental Setup · Derivation · Wavelength & Refractive Index

6.1 Introduction

Newton's Rings are circular interference fringes formed when a plano-convex lens of large radius of curvature R is placed on a flat glass plate. An air wedge of varying thickness is formed between the curved surface of the lens and the flat plate. Monochromatic light falling normally on this arrangement produces concentric alternating bright and dark rings.

6.2 Experimental Setup

1. A plano-convex lens (radius of curvature R) rests on a flat glass plate.
2. Monochromatic light from a sodium lamp falls on a glass plate inclined at 45°, directing light vertically downward.
3. Reflected rays from the top (lens bottom surface) and bottom (glass plate top surface) of the air film interfere.
4. The rings are viewed through a microscope placed vertically above the lens.

6.3 Derivation of Ring Diameter

Let the lens have radius of curvature R, and let r be the radius of any ring where the air gap is t.

1. From geometry of the sphere: $r^2 = R^2 - (R-t)^2 = 2Rt - t^2 \approx 2Rt$ (since $t << R$)
2. Therefore: $t = r^2 / (2R)$
3. Path difference for air film ($\mu = 1$): $2t\cos\theta \approx 2t$ (normal incidence, $\theta = 0$)
   With phase reversal: effective path diff. $= 2t - \lambda/2$
4. Dark ring condition: $2t = n\lambda$
   $\Rightarrow 2 \times r^2/(2R) = n\lambda$
   $\Rightarrow r^2{}_n = nR\lambda$
   $\Rightarrow D_n{}^2 = 4nR\lambda$
5. Bright ring condition: $2t = (2n-1)\lambda/2$
   $\Rightarrow D_n{}^2 = 2(2n-1)R\lambda$

Air gap: $t = r^2 / (2R) = D^2/(8R)$

Dark ring diameter: $D_n^2 = 4nR\lambda \quad (n = 0, 1, 2, \ldots)$

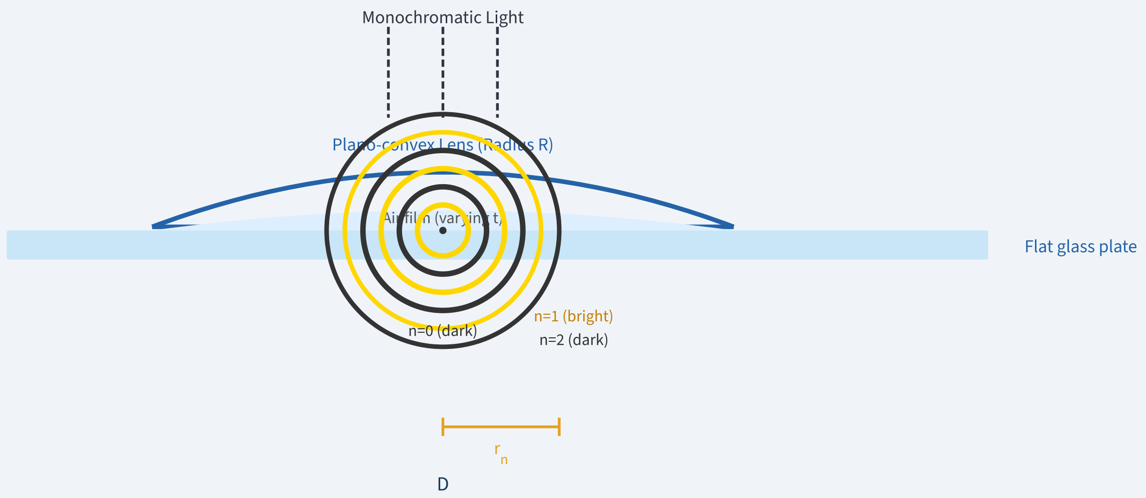
Bright ring diameter: $D_n^2 = 2(2n-1)R\lambda$

Wavelength: $\lambda = (D_{n+p}^2 - D_n^2) / (4pR)$

Refractive index: $\mu = (D_{n+p}^2 - D_n^2)_{air} / (D_{n+p}^2 - D_n^2)_{liquid}$

💡 **NOTE – CENTRE OF NEWTON'S RINGS**

The central spot is always **dark** in reflected light. This is because at the centre t = 0, so path difference = −λ/2 (only from phase reversal), giving destructive interference.



$_n^2 = 4nR\lambda$ (dark) | Centre: dark spot Ray 1 Ray 2

FIG. 6.1 — NEWTON'S RINGS SETUP & RING PATTERN

Concentric rings: gold = bright (odd), dark = destructive. Central spot always dark.

7 Diffraction Overview

Definition · Types · Huygens–Fresnel Principle

7.1 Definition

Diffraction is the bending or spreading of waves around obstacles or apertures. It is a consequence of the wave nature of light. Whenever a wave front encounters an opaque object or an aperture whose dimensions are comparable to the wavelength of light, the wave bends into the geometrical shadow region, producing a diffraction pattern of alternating maxima and minima.

7.2 Huygens–Fresnel Principle

Every point on a wavefront acts as a secondary source of spherical wavelets. The resultant wave at any later time is the envelope of these secondary wavelets, each contributing an amplitude proportional to $(1 + \cos\theta)/2$ in the direction θ. Diffraction patterns are calculated by summing (integrating) contributions from all secondary sources across the aperture, taking amplitude and phase into account.

7.3 Types of Diffraction

Feature	Fresnel Diffraction	Fraunhofer Diffraction
Source distance	Finite (nearby source)	Infinite (or lens used)
Screen distance	Finite (nearby screen)	Infinite (or lens used)
Wavefront at slit	Spherical / cylindrical	Plane
Mathematical treatment	Complex (Fresnel integrals)	Simpler (Fourier transform)
Pattern shape	Resembles geometric shadow	Well-defined maxima/minima
Practical setup	No lenses needed	Converging lenses required

7.4 Comparison: Interference vs Diffraction

Property	Interference	Diffraction
Cause	Superposition of waves from two or more coherent sources	Superposition of secondary wavelets from single wavefront
Fringe width	Equal throughout	Central maximum twice wider; decreasing
Fringe intensity	All bright fringes same intensity	Decreases from centre outward
Minima	Perfectly dark	Not perfectly dark (except single slit)
Number of sources	Two (double slit)	Infinite (continuous wavefront)

8 Fresnel Diffraction

Half-Period Zones · Zone Plate · Qualitative Treatment

8.1 Fresnel's Half-Period Zones (HPZ)

Fresnel divided the wavefront into concentric annular zones such that the distance from successive zone boundaries to the point of observation P differs by $\lambda/2$. The path difference from consecutive zones to P is therefore $\lambda/2$, meaning successive zones are out of phase with each other.

8.2 Construction of Zones

1. Let P be at distance b from the wavefront. Mark point O directly opposite P on the wavefront.
2. Zone 1: all points within distance $b + \lambda/2$ from P (so path = $b + \lambda/2$).
3. Zone m: bounded by distances $b + m\lambda/2$ and $b + (m-1)\lambda/2$ from P.
4. Radius of m-th zone: $r_m = \sqrt{(mb\lambda)}$ (approximately, for $b >> r_m$).

Zone radius: $r_m = \sqrt{(mb\lambda)}$

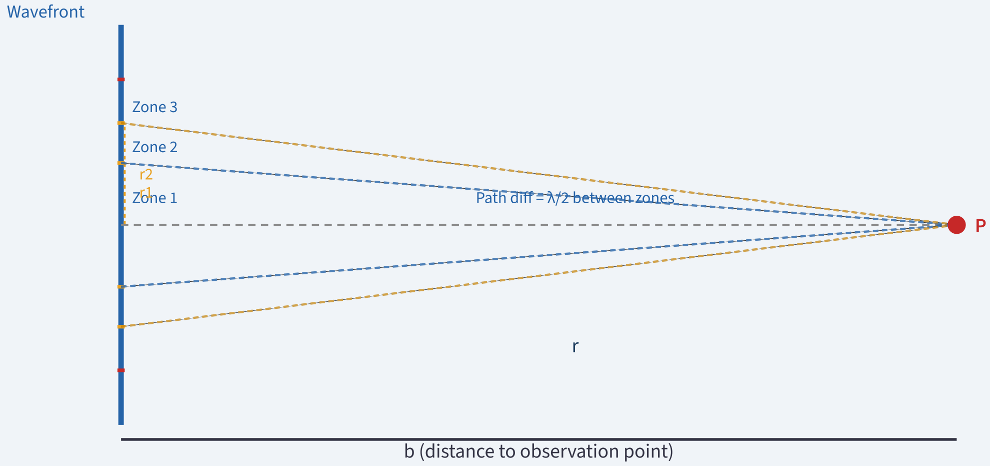
Area of zone m: $A_m \approx \pi b\lambda$ (constant for all zones)

Resultant amplitude: $A = a_1/2$ (half of first zone)

Intensity (unobstructed): $I = I_0$ (normal)

8.3 Zone Plate

A Zone Plate is an optical device that blocks alternate Fresnel zones (either odd or even), so that all remaining zones contribute in phase. This concentrates light at P, making the zone plate act like a converging lens. Focal length: $f = r_m^2/(m\lambda)$.



$_{m} = \sqrt{(mb\lambda)}$ | Zone area ≈ $\pi b\lambda$ (equal)

FIG. 8.1 — FRESNEL HALF-PERIOD ZONES

Each successive zone differs in path by λ/2; zones alternate between constructive and destructive contribution

9

Fraunhofer Diffraction

Single Slit · Double Slit · N-Slit (Qualitative)

9.1 Single Slit Fraunhofer Diffraction

A plane wave is incident normally on a slit of width a. The diffracted light is focused by a lens onto a screen at the focal plane. The intensity distribution is derived by dividing the slit into N infinitesimal strips and summing (integrating) their contributions with appropriate phase differences.

9.2 Derivation of Intensity

1. Consider the slit of width a. For light diffracted at angle θ, the path difference between rays from top and bottom of slit = $a \sin\theta$.
2. Define: $\alpha = (\pi a \sin\theta) / \lambda$
3. The resultant amplitude: $A = A_0 (\sin \alpha) / \alpha$
4. Intensity: $I = I_0 (\sin \alpha / \alpha)^2$
5. Minima occur when $\sin \alpha = 0 \Rightarrow \alpha = \pm m\pi \Rightarrow a \sin\theta = m\lambda$ ($m \neq 0$)
6. Secondary maxima occur approximately at $\alpha = \pm(2m+1)\pi/2$

Parameter: $\alpha = (\pi a \sin\theta) / \lambda$

Intensity: $I(\theta) = I_0 (\sin \alpha / \alpha)^2$

Minima (dark): $a \sin\theta = m\lambda \quad m = \pm1, \pm2, \ldots$

Central maximum: at $\theta = 0$, half-width $= \sin^{-1}(\lambda/a)$

Width of central max.: $2\lambda D/a$ (on screen at distance D)

Subsidiary maxima: $a \sin\theta = (2m+1)\lambda/2$

9.3 Double Slit Diffraction (Qualitative)

For two slits each of width a, separated by d, the pattern is a product of the single-slit diffraction envelope and the two-slit interference pattern. The intensity is:

$$I(\theta) = I_0 \, (\sin \alpha \,/\, \alpha)^2 \times \cos^2\beta$$

where: $\alpha = (\pi a \sin\theta)/\lambda$ and $\beta = (\pi d \sin\theta)/\lambda$

Missing orders: when $a/d = m/n$, order n coincides with diffraction minimum

9.4 N-Slit Diffraction (Qualitative)

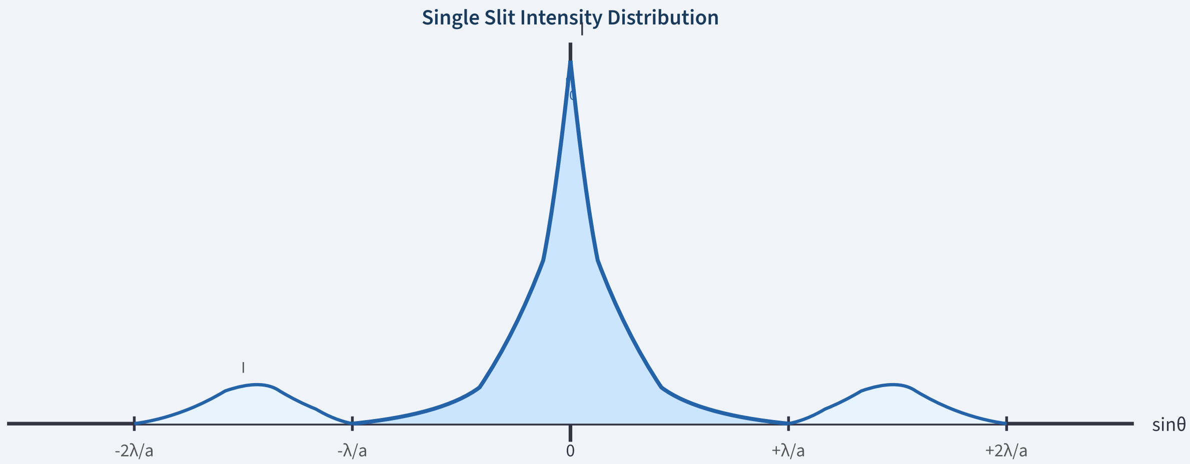
For N equally spaced slits (each width a, spacing d), the intensity is:

$$I(\theta) = I_0 \, (\sin \alpha \,/\, \alpha)^2 \times [\sin(N\beta) \,/\, (N \sin\beta)]^2$$

Principal maxima: $d \sin\theta = m\lambda \quad (N\beta = m\pi)$

Minima between: $N(d \sin\theta) = p\lambda \; (p \neq 0, N, 2N, \ldots)$

Secondary maxima between consecutive principal maxima: $N - 2$



$_0$/22 I_0/22 Central max = 2λD/a wide | $I = I_0(\sin\alpha/\alpha)^2$ Minima at a sinθ = mλ (m = ±1, ±2, …)

FIG. 9.1 — FRAUNHOFER SINGLE SLIT INTENSITY PATTERN

Central maximum is twice as wide as subsidiary maxima; intensity falls as $(\sin\alpha/\alpha)^2$

10 Diffraction Grating

Grating Equation · Dispersive Power · Resolving Power

10.1 Definition

A diffraction grating is an optical device consisting of a large number N of equally spaced parallel slits (or grooves). A transmission grating has transparent slits; a reflection grating has reflecting grooves. A typical grating has 500–5000 lines per mm.

10.2 Grating Equation (Principal Maxima)

1. Let d = grating element = (a + b) = distance between corresponding points on successive slits.
2. For light incident at angle i and diffracted at angle θ, path difference between adjacent slits = d(sin i + sin θ).
3. For constructive interference (principal maxima): d(sin i + sin θ) = mλ.
4. For normal incidence (i = 0): d sinθ = mλ.

> Grating element: `d = 1/N (N = lines per unit length)`
>
> Grating equation: `(a+b) sinθ = mλ    m = 0, 1, 2, …`
>
> Oblique incidence: `(a+b)(sin i + sin θ) = mλ`
>
> Maximum order: m_{max} `= d/λ (when θ = 90°)`

10.3 Dispersive Power (Qualitative)

Dispersive power is the ability of a grating to separate two spectral lines of nearly equal wavelengths. It is defined as:

> Dispersive power: `dθ/dλ = m / (d cosθ)`
>
> Increases with: `order m (higher order → greater dispersion)`
>
> Increases with: `N (more lines → greater dispersion)`
>
> Decreases with: `d (smaller spacing → larger θ → smaller cosθ)`

10.4 Resolving Power (Qualitative)

Resolving power (RP) is the ability of a grating to distinguish between two spectral lines of wavelengths λ and λ+dλ. By Rayleigh's criterion, two lines are just resolved when the principal maximum of one falls on the first minimum of the other.

Resolving power: `RP = λ/dλ = mN`

where: `m = order of diffraction, N = total number of slits`

Increases with: `N (more slits) and m (higher order)`

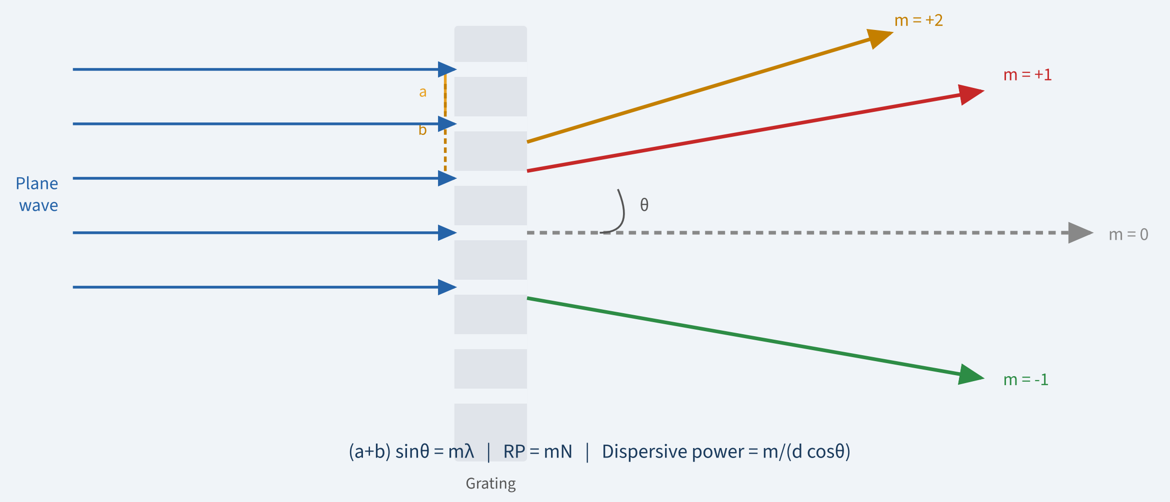


FIG. 10.1 — DIFFRACTION GRATING SETUP

Principal maxima at (a+b)sinθ = mλ. Orders m = 0 (undeviated), ±1, ±2, …

11 Polarization Overview

Types · Brewster's Law · Double Refraction · Malus's Law

11.1 Introduction

Light is a transverse electromagnetic wave; the electric field vector oscillates perpendicular to the direction of propagation. In ordinary (unpolarized) light, the electric field oscillates randomly in all directions perpendicular to propagation. **Polarization** is the process of restricting this oscillation to a specific direction or pattern.

11.2 Types of Polarization

Type	Description	Source
Linear (Plane) Polarized	Electric field oscillates in a single plane containing the propagation direction	Polarizer, Nicol prism
Circular Polarized	Electric field vector rotates uniformly; constant magnitude; traces a circle	Quarter wave plate
Elliptical Polarized	Electric field vector rotates; varies in magnitude; traces an ellipse	Any wave plate
Unpolarized	Electric field oscillates equally in all transverse directions	Ordinary light source

11.3 Polarization by Reflection – Brewster's Law

When unpolarized light strikes a dielectric surface at the Brewster angle (θ_B), the reflected light is completely plane-polarized (vibrations perpendicular to the plane of incidence only). At this angle, the reflected and refracted rays are perpendicular to each other.

Brewster's Law: $\tan\theta_B = \mu = n_2/n_1$

Condition: $\theta_B + \theta_r = 90°$

For glass ($\mu = 1.5$): $\theta_B = \tan^{-1}(1.5) \approx 56.3°$

Reflected ray: completely polarized (s-polarized)

Refracted ray: partially polarized

11.4 Polarization by Refraction (Pile of Plates)

When unpolarized light passes through a stack of glass plates at Brewster's angle, the transmitted light progressively becomes more completely polarized (vibrations in the plane of incidence). With enough plates (about 15–20), the transmitted beam is nearly completely plane-polarized.

11.5 Malus's Law

When plane-polarized light of intensity I_0 passes through an analyser whose transmission axis makes angle θ with the polarization direction:

Malus's Law: $I = I_0 \cos^2\theta$

Maximum: $I = I_0$ when $\theta = 0$

Minimum: $I = 0$ when $\theta = 90°$

11.6 Double Refraction (Birefringence)

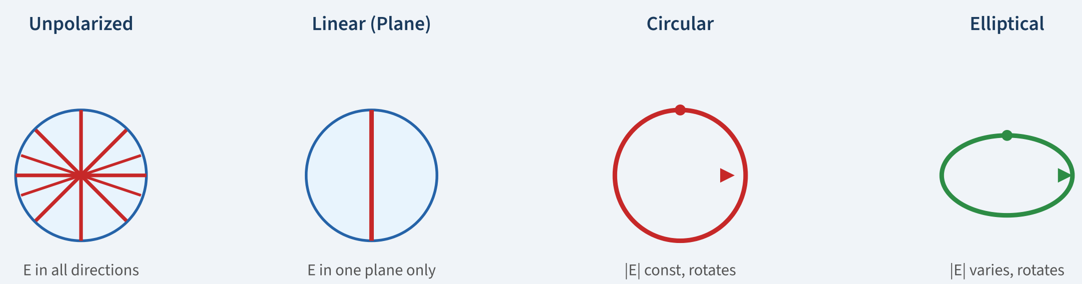
Certain crystals such as calcite (Iceland spar) and quartz split an incident ray into two refracted rays: the **Ordinary ray (O-ray)** and the **Extraordinary ray (E-ray)**. The O-ray obeys Snell's law and has a fixed refractive index (μ_o). The E-ray does not obey Snell's law in general; its refractive index (μ_e) varies with direction. Both rays are plane-polarized in perpendicular directions.

O-ray: always satisfies $\mu_o = \sin i / \sin r$ (same in all directions)

E-ray: $\mu_e(\theta)$ varies between μ_o and μ_e(principal)

Optic axis: direction in crystal along which O and E rays travel at same speed

Calcite: $\mu_o = 1.658$, $\mu_e = 1.486$ (negative uniaxial)



Light propagating into page • E-field configurations shown in cross-section

Malus's Law: I = I

$_0 \cos^2\theta$ | Brewster angle: $\tan\theta_B = \mu$

FIG. 11.1 — STATES OF POLARIZATION

From left: unpolarized, plane-polarized, circularly polarized, elliptically polarized light

12 Optical Devices

Nicol Prism · Half Wave Plate · Quarter Wave Plate

12.1 Nicol Prism

A Nicol Prism is an optical device made from calcite (Iceland spar) that produces plane-polarized light by eliminating one of the two doubly-refracted rays via total internal reflection.

Construction

A calcite crystal of length three times its width is cut along a diagonal and cemented together with Canada Balsam (refractive index $\mu_{CB} = 1.55$, lying between $\mu_o = 1.658$ and $\mu_e = 1.486$ of calcite).

Working Principle

1. Unpolarized light enters the Nicol prism and is split into O-ray ($\mu_o = 1.658$) and E-ray ($\mu_e \approx 1.486$) by double refraction.
2. The O-ray strikes the Canada Balsam layer (denser to rarer medium). Since $\mu_o = 1.658 > \mu_{CB} = 1.55$, the critical angle is $\theta_c = \sin^{-1}(1.55/1.658) \approx 69°$.
3. The O-ray strikes the layer at about 76° which exceeds θ_c, so it undergoes Total Internal Reflection and is absorbed by the blackened side of the prism.
4. The E-ray ($\mu_e < \mu_{CB}$) passes through the balsam layer and emerges as completely plane-polarized light (vibrations in the principal section).

Calcite μ_o: 1.658 (O-ray, sodium light)

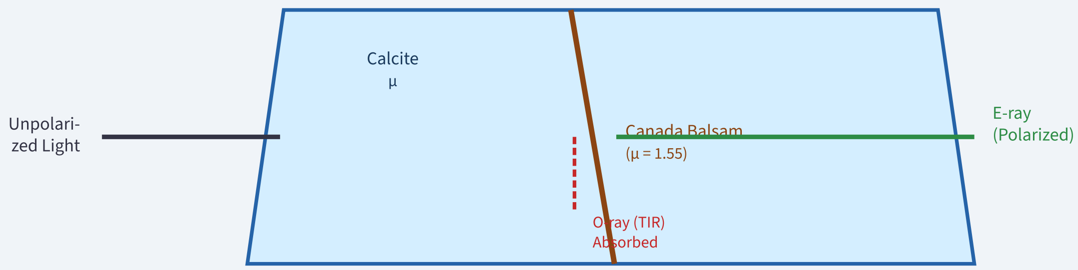
Calcite μ_e: 1.486 (E-ray, sodium light)

Canada Balsam μ: 1.550

Critical angle (O-ray): $\theta_c = \sin^{-1}(1.55/1.658) \approx 69°$

O-ray: Total Internal Reflection → eliminated

E-ray: transmitted as plane-polarized light



$_o$=1.658, μ_e=1.486 Calcite μ_o=1.658, μ_e=1.486 Plane polarized Nicol Prism: O-ray eliminated by TIR; E-ray transmitted as plane-polarized

FIG. 12.1 — NICOL PRISM STRUCTURE & WORKING

O-ray (μ_o=1.658) undergoes TIR at Canada Balsam (μ=1.55); E-ray passes through polarized

12.2 Wave Plates (Retardation Plates)

Wave plates are cut from doubly refracting crystals (e.g., quartz, mica) such that the optic axis is in the plane of the plate. A plane-polarized beam entering the plate is split into O-ray and E-ray which travel in the same direction but with different speeds, creating a phase retardation between them.

Phase Retardation Formula

Phase difference: $\delta = (2\pi/\lambda) \times (\mu_e - \mu_o) \times t$

Half Wave Plate (HWP): $\delta = \pi$ (path diff. $= \lambda/2$)

Thickness (HWP): $t = \lambda / [2(\mu_e - \mu_o)]$

Quarter Wave Plate (QWP): $\delta = \pi/2$ (path diff. $= \lambda/4$)

Thickness (QWP): $t = \lambda / [4(\mu_e - \mu_o)]$

12.3 Half Wave Plate (HWP)

A HWP introduces a phase retardation of π (path difference λ/2) between O and E rays. Effect: If plane-polarized light enters at angle θ to the optic axis, the emergent light is plane-polarized but the plane of polarization is rotated by 2θ. A HWP can convert right-circularly polarized to left-circularly polarized light and vice versa.

12.4 Quarter Wave Plate (QWP)

A QWP introduces a phase retardation of π/2 (path difference λ/4). Effect: If plane-polarized light enters at 45° to the optic axis, the O and E components have equal amplitude and π/2 phase difference, producing circularly polarized light. A QWP can convert plane polarized ↔ circularly polarized, and is used to distinguish circular from linear polarization.

$_{HWP} = \lambda / [2(\mu_e - \mu_o)]$ $t_{QWP} = \lambda / [4(\mu_e - \mu_o)]$ Phase retardation: $\delta = (2\pi/\lambda)(\mu_e - \mu_o)t$

FIG. 12.2 — HALF WAVE PLATE AND QUARTER WAVE PLATE

HWP: rotates plane of polarization by 2θ. QWP at 45°: converts linear ↔ circular polarization.

13

Questions & Answers

2-Mark · 5-Mark · 10-Mark Questions

2-Mark Questions (Short Answer)

2M Q1. Define coherent sources. Why are they necessary for interference?

Coherent Sources are those that emit light waves of the same frequency (or wavelength) and maintain a constant phase difference between them over time. A constant phase difference is essential because interference fringes are formed by the superposition of waves. If the phase difference fluctuates randomly (as with independent sources), the positions of maxima and minima change so rapidly that only a uniform illumination (no fringes) is observed. Real coherent sources are obtained by dividing a single wavefront (wavefront division, e.g., Young's slits) or a single amplitude (amplitude division, e.g., thin films, Newton's rings). Laser sources are highly coherent because of their narrow spectral width and high temporal and spatial coherence.

2M Q2. State and explain Brewster's Law.

Brewster's Law states: The tangent of the polarizing angle (Brewster angle θ_B) equals the refractive index (μ) of the medium: $\tan\theta_B = \mu$.

When unpolarized light strikes a dielectric surface at the Brewster angle, the reflected beam is completely plane-polarized (vibrations only perpendicular to the plane of incidence). At this angle, the reflected and refracted beams are mutually perpendicular ($\theta_B + \theta_r = 90°$). For glass with $\mu = 1.5$, $\theta_B \approx 56.3°$. This law is used in polarimetry, LCD screens, and glare-reduction optics.

2M Q3. Distinguish between Fresnel and Fraunhofer diffraction.

In **Fresnel diffraction**, the source and screen are at finite distances from the diffracting aperture; no lenses are needed, and the incident wavefront is spherical or cylindrical. The mathematical treatment uses Fresnel integrals. In **Fraunhofer diffraction**, the source and screen are at effectively infinite distances (plane waves); converging lenses bring these planes to finite distances in practice. The pattern is the Fourier transform of the aperture function. Fraunhofer patterns are simpler to analyze and show well-defined maxima and minima, while Fresnel patterns resemble the geometric shadow with ripples at the edges.

2M Q4. What is a diffraction grating? Write the grating equation.

A **diffraction grating** is an optical device having a large number of equally-spaced parallel slits (or grooves) ruled on a glass or metal surface. A typical grating may have 500 to 5000 lines per millimetre. The **Grating Equation** for principal maxima is: $(a + b) \sin\theta = m\lambda$, where $(a + b)$ is the grating element (distance between adjacent slits), θ is the angle of diffraction, m is the order of the spectrum (0, 1, 2, …), and λ is the wavelength of light. The grating is used in spectroscopy to measure wavelengths with high accuracy because of its high resolving power $R = mN$.

5-Mark Questions (Medium Answer)

5M Q5. Explain Newton's Rings experiment with diagram. Derive the expression for diameter of dark rings and determine wavelength of light.

Newton's Rings are concentric circular interference fringes formed when a plano-convex lens is placed on a flat glass plate, creating an air film of varying thickness.

Setup: A plano-convex lens (radius of curvature R) rests on a flat glass plate. Monochromatic light from a sodium lamp is reflected by a 45° glass plate to fall normally on the setup. Reflected rays from the top and bottom of the air film interfere. The rings are observed through a travelling microscope.

Derivation: At radius r, air gap $t = r^2/(2R)$. Effective path difference (reflected) $= 2t - \lambda/2$ (phase reversal at denser medium). For dark rings: $2t = n\lambda \Rightarrow r^2/(R) = n\lambda \Rightarrow D_n^2 = 4nR\lambda$.

Measurement of wavelength: Measure diameters D_n and D_{n+p} with a microscope. Then: $\lambda = (D_{n+p}^2 - D_n^2) / (4pR)$. This eliminates the effect of dust or imperfect contact at centre.

Key observations: Centre is dark (dark spot) because $t = 0$ gives path diff. $= \lambda/2$ (purely from phase reversal). Ring spacing decreases outward. In transmitted light, the centre is bright and conditions for bright/dark fringes are reversed.

5M **Q6. Derive the expression for fringe width in Young's Double Slit Experiment.**

Setup: Two coherent slits S_1 and S_2 are separated by distance d. Screen is at distance D. Point P is at height y above centre O.

Path Difference: $S_1P = \sqrt{[D^2 + (y - d/2)^2]}$, $S_2P = \sqrt{[D^2 + (y + d/2)^2]}$. For small y and d: $\Delta = S_2P - S_1P \approx yd/D$.

Bright fringes: $\Delta = n\lambda \Rightarrow y_n = n\lambda D/d$ $(n = 0, 1, 2, \ldots)$
Dark fringes: $\Delta = (2n-1)\lambda/2 \Rightarrow y_n = (2n-1)\lambda D/(2d)$

Fringe width β $= y_{n+1} - y_n = \lambda D/d$ (same for both bright and dark).

Key observations: (i) $\beta \propto \lambda$: fringe width increases with wavelength. (ii) $\beta \propto D$: wider fringes for larger screen distance. (iii) $\beta \propto 1/d$: fringes closer for larger slit separation. (iv) Intensity $I = 4I_0\cos^2(\delta/2)$ where $\delta = (2\pi/\lambda)(yd/D)$.

5M **Q7. Explain interference in thin films by reflection. Write conditions for bright and dark fringes.**

Thin Film Interference (Reflected Light):
When monochromatic light of wavelength λ falls on a thin transparent film of thickness t and refractive index μ at angle of incidence i (refraction angle θ_r):

Ray 1 is reflected at the upper surface (air-to-film interface, going from rarer to denser): undergoes phase change of π (equivalent to extra path λ/2).
Ray 2 refracts into film, reflects at lower surface (film-to-air, denser to rarer): no phase change. Travels extra optical path $2\mu t \cos\theta_r$.

Effective optical path difference: $\Delta_{eff} = 2\mu t \cos\theta_r - \lambda/2$

Bright fringes (constructive): $2\mu t \cos\theta_r = (2n+1)\lambda/2$ $(n = 0, 1, 2, \ldots)$
Dark fringes (destructive): $2\mu t \cos\theta_r = n\lambda$ $(n = 0, 1, 2, \ldots)$

Applications: Anti-reflection coatings on lenses (minimum reflection when $2\mu t = \lambda/2$); high-reflectance multilayer coatings; optical testing for surface flatness using equal-thickness fringes (Fizeau fringes).

5M **Q8. Explain the construction and working of a Nicol Prism.**

Construction: A naturally occurring calcite crystal (length : breadth = 3:1) is cut along a diagonal making 68° with the base. The two halves are cemented back together with Canada Balsam ($\mu_{CB} = 1.55$) which lies between $\mu_o = 1.658$ and $\mu_e = 1.486$ of calcite. The end faces are ground to make 68° with the long axis.

Working:

Step 1: Incident unpolarized light enters and is split by double refraction into O-ray ($\mu_o = 1.658$, vibrations perpendicular to principal section) and E-ray ($\mu_e = 1.486$, vibrations in principal section).

Step 2: O-ray travels from calcite ($\mu = 1.658$) to Canada Balsam ($\mu = 1.55$), a denser-to-rarer transition. Critical angle = $\sin^{-1}(1.55/1.658) \approx 69°$. The O-ray hits at ≈76° > θ_c and undergoes Total Internal Reflection (TIR).

Step 3: TIR O-ray is absorbed by the blackened side.

Step 4: E-ray passes from calcite ($\mu_e < \mu_{CB}$) — rarer-to-denser for E-ray — and transmits through the balsam layer and exits as completely plane-polarized light.

Limitations: Field of view is limited (±14°); not suitable for ultraviolet; Canada Balsam may deteriorate.

5M **Q9. Explain the types of polarization and derive Malus's Law.**

Types of Polarization:

(i) **Plane (Linear) Polarized:** Electric field vector oscillates in a single plane. Produced by polarizers, Nicol prism.

(ii) **Circularly Polarized:** E vector has constant magnitude but rotates with constant angular velocity. Produced by QWP at 45°; equal amplitudes with $\delta = \pi/2$.

(iii) **Elliptically Polarized:** E rotates but magnitude varies, tracing an ellipse. General case: unequal amplitudes or $\delta \neq \pi/2$.

(iv) **Unpolarized:** E oscillates randomly in all transverse directions. Ordinary light sources.

Malus's Law Derivation:

Let plane-polarized light of intensity I_0 (amplitude A_0) pass through an analyser whose axis makes angle θ with the polarization direction. Only the component $A_0\cos\theta$ passes through the analyser. Since intensity ∝ amplitude squared: $I = (A_0\cos\theta)^2 = A_0^2\cos^2\theta = I_0\cos^2\theta$.

Results: $I = I_0$ (max) at $\theta = 0$; $I = 0$ at $\theta = 90°$ (extinction). This is used to verify plane polarization and measure angle of rotation.

5M **Q10. Explain Fraunhofer diffraction at a single slit and derive the intensity distribution.**

Setup: A plane wave falls normally on a slit of width a. A converging lens focuses diffracted light onto a screen at its focal plane.

Derivation:

Divide slit into N infinitesimal strips of width dx. For diffraction angle θ, path difference between element at x and element at 0 is $x \sin\theta$. Phase difference = $(2\pi/\lambda)x \sin\theta$.

Resultant amplitude: $A(\theta) = \int_0^a A_0\, e^{i(2\pi/\lambda)x \sin\theta}\, dx = A_0 a\, (\sin\alpha/\alpha)$ where $\alpha = \pi a \sin\theta/\lambda$.

Intensity: $I(\theta) = I_0(\sin\alpha/\alpha)^2$.

Conditions:

Central maximum at $\theta = 0$ ($I = I_0$).

Minima at $a \sin\theta = m\lambda$ ($m = \pm1, \pm2, \ldots$).

Subsidiary maxima at $a \sin\theta = (2m+1)\lambda/2$ with intensities $I_0/22$, $I_0/62$, …

Width of central max = $2\lambda D/a$ (twice that of subsidiary maxima).

Significance: As slit width a decreases, diffraction increases (wider central maximum). The pattern is the Fourier transform of the slit function.

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Quick Revision Sheet

Key Formulas · Conditions · Diagrams Summary · Important Points

INTERFERENCE – KEY FORMULAS

- Path diff.: $\Delta = yd/D$
- Fringe width: $\beta = \lambda D/d$
- Bright: $\Delta = n\lambda$
- Dark: $\Delta = (2n+1)\lambda/2$
- Intensity: $I = 4I_0\cos^2(\delta/2)$
- $I_{max} = 4I_0$, $I_{min} = 0$

THIN FILMS – REFLECTED LIGHT

- Opt. path diff.: $2\mu t\cos\theta_r$
- Eff. path diff.: $2\mu t\cos\theta_r - \lambda/2$
- Bright: $2\mu t\cos\theta_r = (2n+1)\lambda/2$
- Dark: $2\mu t\cos\theta_r = n\lambda$
- Normal incidence: $\cos\theta_r = 1$

NEWTON'S RINGS

- Air gap: $t = r^2/(2R)$
- Dark rings: $D_n^2 = 4nR\lambda$
- Bright rings: $D_n^2 = 2(2n-1)R\lambda$
- Wavelength: $\lambda = (D_{n+p}^2 - D_n^2)/(4pR)$
- Centre: always dark (reflected)
- RI: $\mu = (D\ \text{air})^2 / (D\ \text{liquid})^2$

DIFFRACTION – SINGLE SLIT

- $\alpha = (\pi a\sin\theta)/\lambda$
- $I = I_0(\sin\alpha/\alpha)^2$
- Minima: $a\sin\theta = m\lambda$
- Width central max: $2\lambda D/a$
- Subsidiary max: $a\sin\theta = (2m+1)\lambda/2$

DIFFRACTION GRATING

- Grating eqn: $(a+b)\sin\theta = m\lambda$
- Max order: $m = d/\lambda$
- Dispersive power: $d\theta/d\lambda = m/(d\cos\theta)$
- Resolving power: $RP = mN$
- N-slit: $I \propto [\sin(N\beta)/(N\sin\beta)]^2$

POLARIZATION

- Malus: $I = I_0\cos^2\theta$
- Brewster: $\tan\theta_B = \mu$
- $\theta_B + \theta_r = 90°$
- HWP: $\delta = \pi$; $t = \lambda/[2(\mu_e - \mu_o)]$
- QWP: $\delta = \pi/2$; $t = \lambda/[4(\mu_e - \mu_o)]$
- Nicol: TIR eliminates O-ray

Summary of Important Derivations

Derivation	Key Steps	Final Result
Fringe width (YDSE)	Path diff. yd/D; bright at $n\lambda$; fringe spacing	$\beta = \lambda D/d$
Thin film (reflected)	Phase reversal at top; OPD = $2\mu t \cos\theta_r$; conditions	Bright: $(2n+1)\lambda/2$
Newton's dark rings	$t = r^2/2R$; $2t = n\lambda$	$D_n^2 = 4nR\lambda$
Single slit intensity	Divide slit; $\alpha = \pi a \sin\theta/\lambda$; integrate	$I = I_0(\sin\alpha/\alpha)^2$
Grating equation	Path diff. between adjacent slits = $d \sin\theta$	$d \sin\theta = m\lambda$
Resolving power	Rayleigh criterion; first min. of one at principal max. of other	RP = mN
Malus's Law	Component through analyser = $A_0\cos\theta$; $I \propto A^2$	$I = I_0\cos^2\theta$
Wave plate retardation	Speed diff. $\mu_e \neq \mu_o$; path diff. = $(\mu_e-\mu_o)t$	$\delta = (2\pi/\lambda)(\mu_e-\mu_o)t$

Memory Tricks

💡 **QUICK MEMORY AIDS**

Thin film conditions (reflected): BRIGHT → ODD (2n+1); DARK → EVEN (2n) in units of λ/2. (Opposite in transmitted light: BRIGHT → nλ; DARK → (2n+1)λ/2.)

Newton's rings centre (reflected): Always DARK — remember "no gap = no path diff., only λ/2 from phase reversal = destructive."

Resolving power: RP = mN — "More lines, More orders, More resolution."

Wave plates: HWP = Half wavelength shift = rotates plane. QWP = Quarter = makes circular.

One-Line Definitions (Exam Ready)

Term	One-line Definition
Interference	Superposition of coherent waves producing alternating bright and dark fringes
Diffraction	Bending of waves around obstacles/apertures due to wave nature of light
Polarization	Restriction of E-field oscillation to a specific direction or pattern
Coherence	Constant phase relationship between two waves over time
Newton's Rings	Concentric circular fringes formed by air wedge between lens and flat plate
Diffraction Grating	Array of equally-spaced slits producing multiple-slit diffraction pattern
Nicol Prism	Calcite device producing plane-polarized light by TIR of O-ray at Canada Balsam
HWP	Wave plate introducing π phase retardation; rotates plane of polarization by 2θ
QWP	Wave plate introducing $\pi/2$ retardation; converts linear to circular polarization at 45°
Brewster angle	Angle of incidence at which reflected light is completely plane-polarized
Fraunhofer diffraction	Far-field diffraction with plane waves; pattern is Fourier transform of aperture
Fresnel diffraction	Near-field diffraction with spherical/cylindrical wavefronts; no lenses required
Double refraction	Splitting of light in anisotropic crystal into O-ray and E-ray
Resolving power	Ability of optical instrument to distinguish two nearby spectral lines

Textbook & Reference Books

Prescribed Texts · Reference Materials

PRESCRIBED TEXTBOOKS

1. *M.N. Avadhanulu, P.G. Kshirsagar & T.V.S. Arun Murthy, A Text Book of Engineering Physics, S. Chand Publishing, New Delhi, 2019.*
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2. *Shatendra Sharma & Jyotsna Sharma, Engineering Physics, Pearson Education India, 2018.*
3. *Sanjay D. Jain, D. Sahasrabudhe & S. Hemne, Engineering Physics, Universities Press, Hyderabad, 2010.*
4. *M.R. Srinivasan, Engineering Physics, New Age International Publishers, New Delhi, 2009.*
5. *Halliday, Resnick & Krane, Physics, Vol. 2, John Wiley & Sons, 5th Ed.*
6. *E. Hecht, Optics, 5th Edition, Pearson Education, 2017.*

⚠ EXAM PREPARATION TIP

For university exams, focus on: (1) Derivation of fringe width for YDSE, (2) Newton's rings diameter formula with diagram, (3) Thin film conditions with stokes relations, (4) Single slit intensity derivation, (5) Grating equation and resolving power, (6) Nicol prism working with TIR explanation, (7) Malus's law derivation. Practice drawing clean, labeled SVG/pencil diagrams for all setup experiments.